Mtazamo Votes Management Portal — Functional Requirements Document

2026-09-19 · @Someone · Version 0.1 (draft for review)

1. Purpose, scope and conventions

This document states the testable functional requirements for the Mtazamo votes management portal. It converts the September 2026 system design and development plan into numbered requirements that the build can be accepted against.

1.1 Scope

The portal is a multi-tenant vote management and parallel tally system for Kenyan aspirants contesting five seats: governor, senator, woman representative, member of parliament and member of county assembly. It supports two independent election segments — party primary/nomination and general election — and covers agent recruitment through to payment, station-level result capture, the tiered verification chain, discrepancy management, war room reporting and evidentiary export. Functionality excluded from this release is listed in section 15.

1.2 Readers

The client and aspirant principals accepting the system, the engineering team building it, QA writing test cases, and any third party auditing the tally in a dispute.

1.3 Conventions

Requirements are numbered `FR-<GROUP>-<n>`: TEN (tenancy), SEAT (seat and tier configuration), GEO (geography), AGT (agents), VOL (volunteers), CAP (capture), VER (verification), DIS (discrepancy), WAR (war room), EXP (exports and payments), SEC (security and audit), NFR (non-functional).

Shall is mandatory. **Should** may be traded against the delivery window with written client agreement. A **hard block** prevents the action from completing. A **soft flag** allows it but records a warning for human attention.

1.4 Definitions

Term	Meaning
Tenant / workspace	An isolated data partition belonging to one aspirant
Segment	One election event — primary or general — with its own geography snapshot, roster, tally and audit trail
Seat type	A configuration record defining a seat's form series, tier count and form ownership per tier
Tier	A level in the collation chain: station, ward, constituency, county
Form A / B / C	Statutory IEBC forms at station, constituency and county level, prefixed by seat series (37A, 37B, 37C for governor)
Verified figure	A station figure a tier officer has confirmed against the photographed form
Discrepancy	A recorded difference between a computed figure and an official figure, or a failed check
Value at risk	The number of votes a given discrepancy could change
Evidence pack	The photograph, typed figures, metadata and audit trail for a single station

2. System overview

The system is a parallel tally: it reconstructs the result independently of the official count, from the same statutory forms, so the campaign knows its position before the declaration and holds the evidence to challenge it afterwards.

Every aspirant gets an isolated workspace holding their own agents, volunteers, geography, forms and tally. No aspirant can see another's data, including where two aspirants contest the same seat in different parties.

The result moves up a chain of custody. Each step adds a human confirmation and leaves an audit record.

```
flowchart TD
    A[Polling station agent<br/>Form A capture] --> B[Ward admin<br/>verify or query]
    B --> C[Constituency agent<br/>Form B reconciliation]
    C --> D[County agent<br/>Form C reconciliation]
    C --> E[War room dashboard]
    D --> E
    B -.query.-> A
    C -.escalation.-> E
```

Governor, senator and woman representative run the full four-tier chain to county. Member of parliament and member of county assembly terminate at the constituency tallying centre, which is where their declaration forms are actually produced.

The system is used once, for roughly eighteen hours, in places with poor connectivity, by people who are tired. Section 14 states the non-functional requirements that follow from that.

3. Actors and access model

Access is derived from geography, never assigned by hand. An agent attached to polling station 042 in Kaptembwo ward automatically inherits the ward admin, the constituency agent and the county agent above them; change the station and the whole chain re-derives.

Level	Who	Scope
Platform admin	Mtazamo staff	All tenants; support and configuration only, never raw tally without client consent
Aspirant / principal	The candidate	Everything in their own workspace
Campaign manager	Aspirant's appointee	Everything except billing and account deletion
Tier officer	County agent, constituency agent, ward admin	Their own geography only, downward

Level	Who	Scope
Polling station agent	Field agent	One polling station, one segment

FR-SEC-1. The system shall derive every user's data scope from their geography assignment at request time, and shall re-derive the full chain automatically when an assignment changes.

FR-SEC-2. The system shall enforce authorisation server-side on every request. Client-side scoping is presentation only and shall never be the sole control.

FR-SEC-3. A platform admin shall be able to perform configuration and support actions without viewing raw tally figures. Access to raw tally by a platform admin shall require recorded client consent and shall be written to the audit log.

FR-SEC-4. No request authenticated against one tenant shall return data belonging to another tenant, under any parameter.

4. Tenancy and election segments (FR-TEN)

The portal runs two independent election segments per workspace.

Segment	What it tallies	Forms used
Party primary / nomination	The party's own nomination exercise	Party-issued tally sheets; no statutory forms. Configurable form template per party
General election	The IEBC-administered national election day	Statutory IEBC forms, by seat (section 5)

FR-TEN-1. The system shall create a workspace per aspirant, scoped to one seat, with all records inside it partitioned by tenant.

FR-TEN-2. Each segment shall be a separate event object carrying its own geography snapshot, agent roster, tally and audit trail. Records in one segment shall not alter records in the other.

FR-TEN-3. An aspirant who wins a primary shall be able to carry the agent roster forward into the general election segment in a single action, without re-recruiting or re-vetting.

FR-TEN-4. An aspirant who loses a primary shall retain full read access and export rights over that segment's data.

FR-TEN-5. The primary segment shall use a configurable form template defined per party, with named result fields, rather than a fixed statutory form.

FR-TEN-6. Workspace creation shall require: seat selection, segment selection, county or counties in scope, and shall land the user on a dashboard that is empty rather than seeded with sample data.

5. Seat, form and tier configuration (FR-SEAT)

Every verification rule keys off this matrix.

Seat	Station form	Ward tier	Constituency form	County form	Chain depth
Governor	37A	Ward admin	37B	37C	Station → ward → constituency → county
Senator	38A	Ward admin	38B	38C	Station → ward → constituency → county
Woman representative	39A	Ward admin	39B	39C	Station → ward → constituency → county
Member of parliament	35A	Ward admin	35B	—	Station → ward → constituency
Member of county assembly	36A	Ward admin	36B	—	Station → ward → constituency

MCA results from all polling stations in a ward are collated at the constituency tallying centre onto Form 36B, with 36C as the declaration form; 35C is the MP declaration form at the constituency. Terminating the MCA and MP chains at a constituency-level B form matches what actually happens at the tallying centre and lets the agent challenge the collation where it is legally challengeable.

FR-SEAT-1. Seat behaviour shall be data-driven, not hard-coded. A `seat_type` record shall carry the seat's form series, its number of tiers, and which tier owns which form.

FR-SEAT-2. Adding a further chain — for example a presidential chain of 34A/34B/34C — shall be a data insert requiring no code change.

FR-SEAT-3. The tier engine shall return, for any seat, the ordered chain of tiers and the form series expected at each tier.

FR-SEAT-4. Tier display names shall be configurable per tenant, so the screen matches the language the campaign uses.

FR-SEAT-5. The capture form shown to an agent shall be generated from the seat's form definition, not from a per-seat hard-coded screen.

6. Geography (FR-GEO)

FR-GEO-1. The system shall import polling stations in bulk from CSV, each record carrying station code and name, ward, constituency, county and registered-voter count.

FR-GEO-2. The importer shall validate on load — unknown ward or constituency codes, duplicate station codes, missing or non-numeric registered voters — and shall report failed rows individually rather than rejecting the whole file.

FR-GEO-3. The system shall roll station-level registered voters up to ward, constituency and county totals automatically.

FR-GEO-4. A station coverage view shall list every station in the tenant's geography with its assigned agent or an explicit unassigned alert.

FR-GEO-5. Each segment shall hold its own geography snapshot, so a boundary or station-list change between primary and general election does not alter the primary's record.

FR-GEO-6. Geography shall be editable after import — a station added, renamed or corrected — with the change written to the audit log.

7. Agent and volunteer management (FR-AGT, FR-VOL)

Recruitment through to payment sits in one module.

7.1 Recruitment and vetting

FR-AGT-1. Each campaign shall have a public application link, shareable by SMS or WhatsApp, collecting name, ID number, phone, ward and preferred station.

FR-AGT-2. Applications shall enter a review queue where the campaign approves or rejects; a rejection shall require a reason.

FR-AGT-3. The system shall verify an applicant's phone by OTP before approval.

FR-AGT-4. The system shall detect duplicate national ID and duplicate phone across the tenant. The same national ID shall not be an agent at two stations or draw two payments.

7.2 Assignment and readiness

FR-AGT-5. The campaign shall be able to assign agents to stations individually, by drag, or in bulk by ward.

FR-AGT-6. The system shall flag every unassigned station in a ward so no station is left uncovered.

FR-AGT-7. Reserve agents shall sit in a bench pool per ward and shall be swappable into a station from the dashboard in one action.

FR-AGT-8. Each agent shall have a readiness checklist — walkthrough watched, test upload completed, attendance confirmed — and the campaign shall see a readiness board by ward showing the percentage of coverage actually confirmed.

7.3 Volunteers

FR-VOL-1. A volunteer record shall be lighter than an agent record: no station assignment and no verification rights.

FR-VOL-2. Volunteers shall be grouped by ward and tagged by activity — mobilisation, transport, door-to-door, social.

FR-VOL-3. Volunteers shall be assignable tasks with a due date and shall check in by phone.

FR-VOL-4. The volunteer board shall report against the turnout target per ward, not against tasks completed.

7.4 Payment

FR-AGT-9. The system shall produce a payment schedule by ward listing agent, phone, station covered, submissions made and amount.

FR-AGT-10. The schedule shall export as a CSV formatted for bulk M-PESA disbursement.

FR-AGT-11. An agent who made no submission shall carry a hold flag on the schedule.

FR-AGT-12. The system shall record the disbursement reference against the agent, so the campaign can answer whether a named agent was paid.

8. Station capture (FR-CAP)

8.1 Result entry

FR-CAP-1. An authenticated agent shall land directly on their assigned station, with no station selection step.

FR-CAP-2. The capture form shall present candidate rows from the seat's form definition plus rejected ballots, disputed ballots, total valid votes and registered voters.

FR-CAP-3. The system shall validate arithmetic at the point of capture:

- Votes cast shall not exceed registered voters — hard block.
- Candidate votes plus rejected ballots shall equal total cast — hard block.
- Turnout above 95%, or any candidate at 100% — soft warning, submission allowed.

FR-CAP-4. A hard block shall state which figures conflict, and shall not be overridable by the agent.

FR-CAP-5. Only one submission per agent per station per segment shall be accepted. A correction shall create a new version, never overwrite.

8.2 Image capture

FR-CAP-6. The agent shall photograph the signed statutory form through the app, with retake, and with multi-page capture for long forms.

FR-CAP-7. Images shall be compressed client-side to roughly 300–500 KB before upload, with the original retained on the device.

FR-CAP-8. Uploads shall use signed URLs, be virus- and file-type checked, be stored under tenant-scoped paths, and generate a thumbnail.

FR-CAP-9. A submission shall not be considered complete for verification until at least one legible form image is attached.

8.3 Metadata

FR-CAP-10. Every submission shall carry GPS coordinates, device fingerprint, timestamp and agent identity.

FR-CAP-11. A submission whose GPS falls outside a configurable radius of the registered station shall be flagged, not blocked. Rural GPS drift is expected and agents shall not be locked out on election night.

8.4 Offline queue

FR-CAP-12. The agent app shall be an installable PWA that captures results and images with no network connection.

FR-CAP-13. Queued submissions shall sync automatically when signal returns, oldest first, with conflict handling on resubmission.

FR-CAP-14. The agent shall see their own queue state — pending, sending, delivered, failed — at all times.

8.5 OCR assist

FR-CAP-15. Uploaded images shall be passed to OCR on a queue, asynchronously, never blocking submission.

FR-CAP-16. Extracted figures shall be displayed beside the agent's typed figures, never in place of them, with mismatches highlighted.

FR-CAP-17. The typed figure shall remain the authoritative value. OCR output shall have no effect on the tally.

8.6 SMS fallback

FR-CAP-18. The system shall accept a structured inbound SMS carrying the numeric result where data connectivity fails entirely.

FR-CAP-19. An SMS submission shall be parsed, matched to its station and agent, and marked as awaiting photograph until the image arrives.

FR-CAP-20. A malformed SMS shall trigger a reply stating the expected format.

9. Verification chain (FR-VER)

9.1 Ward verification

FR-VER-1. The ward admin shall see a live board of every station in their ward with its state: submitted, pending, flagged.

FR-VER-2. Opening a submission shall display the photographed form and the typed figures side by side.

FR-VER-3. The ward admin shall be able to **verify** or **query** a submission. A query shall return it to the agent with a required reason and a notification.

FR-VER-4. The ward admin shall not be able to sign off the ward until every station is either verified or formally marked non-reporting with a reason.

9.2 Constituency reconciliation

FR-VER-5. The constituency agent shall receive verified ward totals rolled up automatically.

FR-VER-6. The constituency agent shall photograph and upload the official Form B and enter its declared figures.

FR-VER-7. The system shall reconcile the verified sum against the official Form B per candidate, and per station where the form is itemised.

FR-VER-8. Any variance shall be displayed as a discrepancy row stating magnitude and direction.

9.3 County reconciliation

FR-VER-9. For governor, senator and woman representative, the county agent shall repeat the exercise one tier up: verified constituency totals against the official Form C, using the same reconciliation engine.

FR-VER-10. For member of parliament and member of county assembly, the chain shall terminate at the constituency with no county step.

9.4 Manual entry and overrides

FR-VER-11. An officer shall be able to add a manual figure with a required reason code from a fixed list: missing station, altered figure, unsigned form, refused access, arithmetic error on the official form.

FR-VER-12. A manual entry shall never overwrite the computed figure. It shall sit alongside it with an audit record of who entered it and when.

FR-VER-13. Both the computed and the manual figure shall remain visible and exportable at every tier above.

10. Discrepancy engine (FR-DIS)

Three classes of check run at every tier.

Class	Check	Behaviour
Arithmetic	Candidate votes + rejected = total cast; cast ⩽ registered	Hard block at capture
Statistical	Turnout outlier, single-candidate sweep, station-level swing far from ward mean	Soft flag, ranked by severity
Reconciliation	Verified sum vs official B or C form	Discrepancy row; blocks tier sign-off until resolved or formally noted

FR-DIS-1. Every discrepancy shall carry a status and a value at risk in votes.

FR-DIS-2. Status shall follow the lifecycle below, with each transition recording actor, timestamp and reason.

```
stateDiagram-v2
    [*] --> Open
    Open --> Explained
    Open --> Escalated
    Explained --> Resolved
    Escalated --> Resolved
    Escalated --> Explained
    Resolved --> [*]
```

FR-DIS-3. Escalation shall be upward and automatic. A discrepancy the ward admin cannot resolve shall become visible to the constituency agent, then the county agent, then the aspirant, each showing elapsed time at that tier.

FR-DIS-4. The war room shall rank open discrepancies by value at risk, so attention goes to what could change the result rather than to a three-vote transcription error in a station of 200 voters.

FR-DIS-5. No tier shall be able to sign off with an unaddressed discrepancy. Formally noting a discrepancy with a reason shall count as addressing it; silently ignoring it shall not.

FR-DIS-6. Statistical thresholds — turnout ceiling, swing tolerance — shall be configurable per tenant without a code change.

FR-DIS-7. Every discrepancy shall link through to the underlying submission, its photograph and its audit trail.

11. War room (FR-WAR)

FR-WAR-1. The aspirant's dashboard shall show, auto-refreshing: stations reporting against total, running verified tally, turnout by ward, and a leader board by candidate.

FR-WAR-2. The dashboard shall show a projected final position, computed by linear projection using registered voters in stations not yet reporting, with a confidence band.

FR-WAR-3. The projection shall be labelled as a projection wherever it appears, and shall never be presented in the same figure as the verified tally.

FR-WAR-4. A discrepancy centre shall list open discrepancies ranked by value at risk, each drilling through to the photograph and the audit trail.

FR-WAR-5. A map view shall drill from county to constituency to ward to station, coloured by reporting status and by margin.

FR-WAR-6. Every figure on the dashboard shall trace back to a photograph and a named agent in no more than two steps.

FR-WAR-7. The dashboard shall distinguish verified figures from submitted-but-unverified figures, and shall never silently include the latter in the headline tally.

12. Exports and notifications (FR-EXP)

FR-EXP-1. The system shall export the full tally to CSV and PDF at any tier.

FR-EXP-2. The system shall produce a per-station evidence pack containing the photograph, typed figures, capture metadata, every verification action and the audit trail.

FR-EXP-3. An evidence pack for any single station shall be producible in no more than two clicks from the dashboard.

FR-EXP-4. The system shall export the audit log in full for a dispute file, in a format readable outside the platform.

FR-EXP-5. The system shall export the agent payment CSV described in FR-AGT-10.

FR-EXP-6. The system shall send SMS and push notifications on: a query raised against an agent's submission, a discrepancy escalation, a tier sign-off, and a station going silent past a configurable threshold.

FR-EXP-7. Notification recipients and thresholds shall be configurable per tenant.

13. Security, audit and compliance (FR-SEC continued)

The audit trail is what makes the tally usable if the campaign goes to a dispute tribunal. It is a functional requirement, not an operational nicety.

FR-SEC-5. Every record shall be append-only. A correction shall create a new version carrying actor, timestamp and reason. Nothing shall be deleted.

FR-SEC-6. The audit log shall cover every verification, override and sign-off, and shall be exportable in full.

FR-SEC-7. Images shall be served through signed, expiring URLs. No storage bucket shall be publicly readable.

FR-SEC-8. Data shall be encrypted in transit and at rest.

FR-SEC-9. Agent personal data shall be handled under the Data Protection Act, 2019, with a stated retention period and an automated deletion routine that runs after the petition window closes.

FR-SEC-10. Two-factor authentication shall be mandatory for tier officers and the aspirant. Field agents shall log in by OTP and shall not be required to manage passwords.

FR-SEC-11. Rate limiting and abuse controls shall apply per tenant. A compromised agent account shall not be able to flood the tally.

FR-SEC-12. Cross-tenant access attempts shall be logged and alerted on, not merely refused.

14. Non-functional requirements (FR-NFR)

14.1 Load

Traffic is not smooth. Expect 60–70% of all submissions inside a two-hour window after the count closes. A county governor race runs roughly 300 to 2,500 polling stations depending on the county, each uploading one image and one form.

FR-NFR-1. The system shall sustain 2,500 stations submitting inside two hours, image upload included, without loss.

FR-NFR-2. Ingestion shall be queue-based, not synchronous. Degradation under peak shall be graceful: submissions queue and land late rather than fail.

FR-NFR-3. The queue and image pipeline shall be sized for the peak, not the average.

14.2 Connectivity and device

FR-NFR-4. Full capture — figures and photograph — shall work with no network connection.

FR-NFR-5. A submission captured in airplane mode shall reach the ward admin's board within seconds of signal returning.

FR-NFR-6. The upload path shall function over a 2G link. A 4 MB photograph shall not be uploaded; see FR-CAP-7.

14.3 Usability

FR-NFR-7. The agent interface shall use large fields, one screen per form section, and no horizontal scrolling.

FR-NFR-8. The agent interface shall be available in Swahili and English, switchable by the agent.

FR-NFR-9. An agent shall be able to complete a station capture without prior training beyond the in-app walkthrough.

14.4 Availability

FR-NFR-10. The system shall be monitored with alerting, with an on-call rota published for election night and a tested rollback plan.

FR-NFR-11. Backups shall run on the production environment with a documented restore procedure.

15. Out of scope

The following are explicitly excluded from this release. Any of them entering scope displaces a listed requirement, in writing.

Excluded	Note
Authoritative OCR extraction	OCR is assistive only (FR-CAP-17). A trustworthy pipeline for handwritten statutory forms is a multi-month problem
Native iOS and Android apps	The agent app is a PWA

Excluded	Note
Biometric agent verification	Vetting is ID plus phone OTP
In-app M-PESA disbursement	CSV export for bulk disbursement instead (FR-AGT-10)
Public results microsite	Tally is visible inside the tenant only
Multi-county aggregation for a presidential chain	The data model supports it (FR-SEAT-2); the screens are not built
Advanced projection modelling	Linear register-based projection only (FR-WAR-2)
In-app chat	Notifications only (FR-EXP-6)

16. Assumptions, dependencies and open decisions

16.1 Assumptions

1. Scope is frozen on day one. Any addition displaces something on the in-scope list.
2. The team is at least six people working in parallel from day one, with geography data and design decisions already made.
3. OCR is assistive, not authoritative.
4. If the system is to be used for a real election night, a full-scale field rehearsal happens at least two weeks before the day. The build fits two weeks; the confidence does not.

16.2 External dependencies

Dependency	Needed for	Owner
Polling-station list with ward, constituency, county codes and registered-voter counts	FR-GEO-1, and the whole verification chain	Client
SMS gateway credentials	FR-CAP-18, FR-EXP-6	Client / vendor
Cloud accounts, domains, object storage, error tracking	Deployment	Build team

Dependency	Needed for	Owner
Party nomination form template	FR-TEN-5	Client, per party

16.3 Open decisions blocking the build

- [ ] Confirm the seat and form matrix, including whether 35C, 36B and 36C are in scope as described in section 5.
- [ ] Confirm the tier names the client wants displayed on screen (FR-SEAT-4).
- [ ] Confirm the primary-segment form template fields (FR-TEN-5).
- [ ] Sign off the out-of-scope list in section 15.
- [ ] Confirm the county or counties in scope for the pilot.

17. Acceptance and traceability

A requirement group is accepted when its exit criterion below is demonstrated to the client and signed against, not when the code exists.

Milestone	Requirement groups	Exit criterion
M1 Foundations (days 1–2)	FR-TEN, FR-SEAT, FR-SEC-1 to 4	A governor tenant and an MCA tenant can be created; the tier engine returns the correct chain and form series for each; a user assigned to a station sees only that station
M2 People (days 3–5)	FR-GEO, FR-AGT-1 to 8, FR-VOL	A governor campaign can recruit, vet, assign and see readiness across every station in one county, and every station shows either an agent or an alert

Milestone	Requirement groups	Exit criterion
M3 Capture (days 5–8)	FR-CAP	An agent in airplane mode can capture a Form 37A, queue it, regain signal and have it appear on the ward admin's board within seconds, with OCR figures shown alongside
M4 Verification (days 8–10)	FR-VER, FR-DIS	A seeded county with deliberately planted errors produces the exact expected discrepancy list at ward, constituency and county tiers, and no tier can sign off with an unaddressed discrepancy
M5 War room and outputs (days 10–12)	FR-WAR, FR-EXP, FR-AGT-9 to 12	The aspirant's dashboard reconciles exactly with the underlying submissions, and the evidence pack for any single station can be produced in two clicks
M6 Hardening (days 12–14)	FR-NFR, FR-SEC-5 to 12	Load test at 2,500 concurrent stations passes; a full election-night drill completes with no critical defects; production is live and monitored

M1 to M4 alone constitute a usable tally system. M5 and M6 enhance it. If the window proves too tight, that is the order in which the system degrades.

Definition of done

A requirement is done when it is merged to trunk, deployed to staging, covered by tests for the verification and discrepancy logic, demonstrated to the client, and accepted against its exit criterion in writing.